



RUNBOOK

## VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

# SDDC Manager Resource Locks API Inspection & Release

Identify, diagnose, and clear stale resource locks that block credential and lifecycle operations — via the REST API, with the verified escalation path.

SDDC MANAGER

RESOURCE LOCKS

REST API

OPERATIONS\_MANAGER

PASSWORD REMEDIATE

KB 432285

RUNBOOK

**v1.0**

VMware Cloud Foundation 9.0.1

# SDDC Manager Resource Locks — API Inspection & Release

## Document Information

|               |                                                                                                                                                                  |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Document Type | Operational Runbook                                                                                                                                              |
| Author        | Virtual Control LLC                                                                                                                                              |
| Issued        | August 5, 2026                                                                                                                                                   |
| Version       | 1.0                                                                                                                                                              |
| Platform      | VMware Cloud Foundation 9.0.1 (SDDC Manager)                                                                                                                     |
| Scope         | Inspecting and clearing stale/orphaned SDDC Manager resource locks that block credential (password rotate/update/remediate) and lifecycle workflows              |
| Verification  | All API behaviors in Sections 4–7 were exercised live against a running 9.0.1 SDDC Manager; the release-method finding (Section 7) is a verified negative result |

**The one-sentence summary:** On VCF 9.0.1 the REST API can **list and diagnose** resource locks ( `GET /v1/resource-locks` ) but **cannot delete them** ( `DELETE` returns HTTP 400 — "method not supported"); clearing a stale lock therefore requires the **KB 432285 service restart**, with a platform-DB delete as the last resort.

## 1. Purpose & Scope

This runbook covers, in order:

- **Authenticating** to the SDDC Manager REST API.
- **Listing** current resource locks and reading the fields that matter.
- **Diagnosing** which workflow owns a lock, and whether that workflow is dead (making the lock stale/orphaned).
- **Attempting** the API release — and the verified result that this is **not supported** on 9.0.1.
- **Escalating** to the Broadcom-prescribed service restart (KB 432285), and the DB-level cleanup as last resort.
- **Verifying** the locks are gone and resuming the blocked operation.

It applies whenever a credential or lifecycle action fails with a lock or "active workflow" error, most commonly after a **failed password rotate/remediate**.

## 2. Symptoms — when to use this runbook

- A password **Rotate / Update / Remediate** fails in the VCF Operations / SDDC Manager UI with "**Unable to acquire resource level lock(s) for workflow id ...**".
- "**Found active workflow for OPERATIONS\_MANAGER**" on a new operation.
- SDDC Manager **health check** reports resource-lock errors.
- The License / Password Management screen shows a lingering red banner referencing a **failed remediate** for a specific resource and user.
- Multiple locks appear on `domain`, `system`, `nsx`, and `vcenter` at once — the signature of a single failed credential workflow that grabbed them all and never released them.

### 3. Prerequisites & the mandatory gate

**Hard gate — 0 in-progress tasks.** Do **not** clear locks (by any method) while a workflow is genuinely running. Confirm zero active tasks first (Section 5.3). Clearing a lock out from under a live workflow corrupts its state.

- Network reachability to `sddc-manager.lab.local` (443).
- SDDC Manager API account (local `admin@local`) and, for escalation, SSH access as the `vcf` user.
- `curl` and (optionally) `jq` / `python` for parsing JSON.
- **Never** run "**Rotate All Passwords**" or **retry the failed remediate** while diagnosing — that re-jams the locks.

### 4. Conventions — how to run these commands

**Where to type these:** a **Linux/macOS shell**, **Git Bash**, or the SDDC Manager appliance shell. They are written for `bash`.

**PowerShell users:** the built-in `curl` is an alias for `Invoke-WebRequest` and will **not** parse these flags. Use `curl.exe` explicitly, and be aware that JSON bodies need escaped quotes in PowerShell. When in doubt, run these from Git Bash.

**Copy the whole grey box.** Commands use `\` line-continuations and a `$TOK` variable carried between steps — partial copies break. Replace `<SDDC_ADMIN_PW>` with the real `admin@local` password (single-quote it so shell metacharacters like `!` are literal).

## 5. Procedure — Inspect & Diagnose (API, read-only)

### 5.1 Acquire an API token

```
TOK=$(curl -sk -X POST "https://sddc-manager.lab.local/v1/tokens" \
-H "Content-Type: application/json" \
-d '{"username":"admin@local","password":"<SDDC_ADMIN_PW>"}' \
```

```
| sed -n 's/.*"accessToken": "\([^"]*\)".*/\1/p' )
echo "token length: ${#TOK}"
```

**Expect:** a token length in the low thousands ( $\approx 2200$  chars). A length of 0 means the credentials or hostname are wrong.

## 5.2 List the current resource locks

```
curl -sk -H "Authorization: Bearer $TOK" -H "Accept: application/json" \
  "https://sddc-manager.lab.local/v1/resource-locks" | python -m json.tool
```

**Expect** an `elements` array; each lock has these fields, all of which matter:

| Field                                                 | Meaning / why it matters                                                                                                                        |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>id</code>                                       | The lock's own UUID (would be the delete target — see Section 7).                                                                               |
| <code>resourceType</code> / <code>resourceName</code> | What is locked: <code>domain</code> , <code>system</code> , <code>nsx</code> , <code>vcenter</code> , etc.                                      |
| <code>operationId</code>                              | <b>Key diagnostic.</b> The workflow that acquired the lock. Locks that share one <code>operationId</code> were all taken by a single operation. |
| <code>serviceId</code>                                | The service that owns it, e.g. <code>OPERATIONS_MANAGER</code> (credential/password operations).                                                |

**Diagnostic tell:** if several locks share one `operationId` and `serviceId = OPERATIONS_MANAGER`, they were all grabbed by a **single credential workflow**. If that workflow is no longer running (Section 5.3), the locks are **orphaned** and safe to clear.

## 5.3 Confirm the owning workflow is dead (and the gate is met)

List recent tasks and look for the matching failed operation and for any still-active task:

```
curl -sk -H "Authorization: Bearer $TOK" -H "Accept: application/json" \
  "https://sddc-manager.lab.local/v1/tasks?limit=50" \
  | python -c "import sys,json
d=json.load(sys.stdin); els=d.get('elements',[])
act=[e for e in els if e.get('status') in ('IN_PROGRESS','PENDING','Running')]
print('ACTIVE tasks:', len(act))
for e in act: print(' -', e.get('status'), '|', e.get('name'))
print('recent FAILED credential ops:')
for e in els:
    if e.get('status') in ('FAILED','Failed') and 'redential' in (e.get('name') or ''):
        print(' -', e.get('creationTimestamp'), '|', e.get('name'))"
```

**Decision:**

- **ACTIVE tasks > 0** → **STOP**. Wait for them to finish (or fail). Re-run until it reads 0.

- **ACTIVE tasks = 0** and a **FAILED "Credentials remediate/rotate operation"** appears → the locks are orphaned by that failure. Proceed.

## 6. What NOT to do

**Do not "Rotate All Passwords."** It attempts every managed account — including whatever account caused the original failure — and re-acquires the same locks.

**Do not retry the failed Remediate** against a resource whose service account is missing/mismatched. Remediate on a missing account is a common way to **jam these locks in the first place** (it takes `domain + system + nsx + vcenter` and dies). Fix the underlying account first; clear the locks second.

## 7. Attempt the API release — and the verified 9.0.1 result

The intuitive step is to delete each lock by its `id` :

```
for id in <lockId-1> <lockId-2> <lockId-3> <lockId-4>; do
  curl -sk -o /dev/null -w "DELETE $id -> HTTP %{http_code}\n" \
    -X DELETE -H "Authorization: Bearer $TOK" \
    "https://sddc-manager.lab.local/v1/resource-locks/$id"
done
```

**Verified result (VCF 9.0.1):** every delete returns **HTTP 400** with `REST_INVALID_API_INPUT / ANNOTATIONS_MISMATCH` and the message **"Request method 'DELETE' is not supported."** The lock count is unchanged.

**Conclusion:** on 9.0.1 the resource-locks endpoint is **read-only** — the API can inspect but not release locks. Do not keep retrying the delete; move to Section 8.

Run the delete anyway as a one-time confirmation on your build/version — if a future release re-enables it, a `200/204` here would clear the lock and you can skip Section 8. As of 9.0.1 it does not.

## 8. Escalation A — Service restart (Broadcom-prescribed, KB 432285)

This is the correct clearing method on 9.0.1. Restarting the SDDC Manager services refreshes `operationsmanager` and `commonsvcs` , which drops the orphaned lock entries.

**Impact:** the SDDC Manager management plane (its UI/API) is unavailable for roughly **3–5 minutes** during the restart. **No impact** to vCenter, NSX, vSAN, or running workloads. Confirm the Section 5.3 gate (0 active tasks) first.

#### Steps:

1. SSH to the SDDC Manager appliance as the `vcf` user:

```
ssh vcf@sddc-manager.lab.local
```

2. Run the service-restart script:

```
/opt/vmware/vcf/operationsmanager/scripts/cli/sddcmanager_restart_services.sh
```

3. Wait for the services to report healthy (the script reports as it restarts each service), then re-authenticate (Section 5.1) and re-list the locks (Section 5.2).

**Expect:** `elements` is now empty — **0 locks**.

## 9. Escalation B — Platform-DB delete (last resort)

If the service restart does **not** clear a genuinely orphaned lock, remove the lock rows directly from the SDDC Manager platform database.

**Last resort only.** Direct DB edits are unsupported for routine use and require care with trust-auth / `pg_hba` and exact row targeting. Use the dedicated procedure — do not improvise SQL against the platform DB.

- Follow the companion runbook: **RUNBOOK-SDDC-Manager-Resource-Lock-DB-Cleanup (v1.2, 2026-06-09)**, which documents the verified platform-DB connection and the targeted `DELETE` against the lock table.
- Same gate applies: **0 in-progress tasks**.

## 10. Verify & resume

1. Re-list locks (Section 5.2) → confirm **0**.
2. Re-check tasks (Section 5.3) → confirm **0 active**, no new FAILED lock/credential task.
3. Resume the originally blocked operation — e.g. the **vCenter root password Update** (VCF Operations → Fleet Management → Passwords), or the corrected credential remediate once the underlying account is fixed.

## 11. Worked example (2026-08-05)

| Step            | Observed                                                                                                                                                                           |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Trigger         | vCenter root-password change was blocked; License Management showed a red banner: "Password remediate for resource: esxi04.lab.local, user: svc-vcf-esxi01, credential type: SSH." |
| Inspect (5.2)   | 4 locks: domain , system , nsx , vcenter — <b>all sharing one</b> operationId 6294f9c1... , serviceId OPERATIONS_MANAGER .                                                         |
| Diagnose (5.3)  | 0 active tasks; task history showed " <b>Credentials remediate operation</b> " <b>FAILED</b> — the orphaning workflow. Locks confirmed stale.                                      |
| API release (7) | All 4 DELETE calls returned <b>HTTP 400</b> — " <b>method not supported.</b> " Locks unchanged. API is read-only on 9.0.1.                                                         |
| Resolution      | Escalated to KB 432285 service restart (Section 8). Root account fix (esxi04 svc-vcf-esxi01 ) handled separately so Remediate is not retried into the same jam.                    |

## 12. Reference

| Source             | Topic                                                                                                     |
|--------------------|-----------------------------------------------------------------------------------------------------------|
| Broadcom KB 432285 | Password remediation fails — "Unable to acquire resource level lock(s)"; service-restart clearing method  |
| Broadcom KB 441213 | "Found active workflow for OPERATIONS_MANAGER"                                                            |
| Broadcom KB 439473 | SDDC Manager health check fails with resource-lock errors                                                 |
| Broadcom KB 416351 | ESXi host service-account credential mismatch (common source of the failed remediate that jams the locks) |
| VC LLC Runbook     | RUNBOOK-SDDC-Manager-Resource-Lock-DB-Cleanup (v1.2, 2026-06-09) — Escalation B                           |

## 13. Revision History

| Version | Date           | Author              | Change                                                                                                                                                                                                                                                                                                                                 |
|---------|----------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | August 5, 2026 | Virtual Control LLC | Initial release. Documents the API inspection/diagnosis workflow, the verified negative result that DELETE /v1/resource-locks/{id} is unsupported on VCF 9.0.1 (HTTP 400), and the KB 432285 service-restart / DB-cleanup escalation path. Includes the 2026-08-05 worked example (4 locks orphaned by a failed esxi04 SSH remediate). |